

11. Remote Support

SSH = Secure Shell

SSH is a predecessor to TELNET, TELNET was not encrypted, but SSH is encrypted.

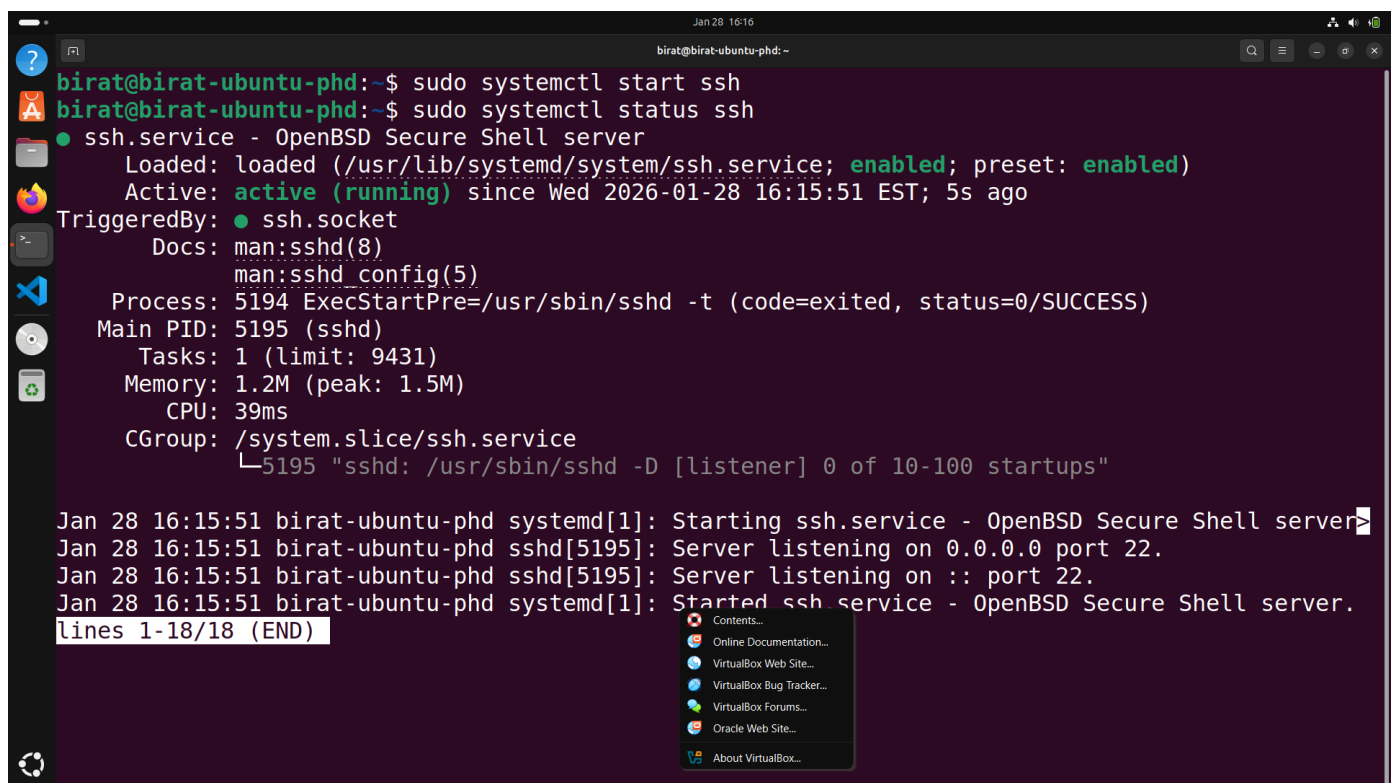
set up ubuntu as the SSH server
and set up win 11 as the SSH client.

sudo apt install openssh-server = installs the ssh server

sudo systemctl enable ssh = enables ssh

sudo systemctl start ssh = starts ssh

sudo systemctl status ssh = shows ssh status

A terminal window screenshot from a Linux system. The user runs 'sudo systemctl start ssh' and 'sudo systemctl status ssh'. The status output shows 'ssh.service - OpenBSD Secure Shell server' is loaded and active (running) since Wed 2026-01-28 16:15:51 EST. It lists the process as 'sshd' with PID 5195, listening on port 22. Below the status, system logs show the service starting and listening on 0.0.0.0 and :: port 22. A context menu is open over the logs, showing options like 'Contents...', 'Online Documentation...', 'VirtualBox Web Site...', 'VirtualBox Bug Tracker...', 'VirtualBox Forums...', 'Oracle Web Site...', and 'About VirtualBox...'.

```
Jan 28 16:16
birat@birat-ubuntu-phd: ~
birat@birat-ubuntu-phd:~$ sudo systemctl start ssh
birat@birat-ubuntu-phd:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-01-28 16:15:51 EST; 5s ago
     TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 5194 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 5195 (sshd)
      Tasks: 1 (limit: 9431)
     Memory: 1.2M (peak: 1.5M)
        CPU: 39ms
     CGroup: /system.slice/ssh.service
            └─5195 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 28 16:15:51 birat-ubuntu-phd systemd[1]: Starting ssh.service - OpenBSD Secure Shell server.
Jan 28 16:15:51 birat-ubuntu-phd sshd[5195]: Server listening on 0.0.0.0 port 22.
Jan 28 16:15:51 birat-ubuntu-phd sshd[5195]: Server listening on :: port 22.
Jan 28 16:15:51 birat-ubuntu-phd systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
lines 1-18/18 (END)
```

default ssh port 22 is open.

edit the firewall rules to allow the ssh traffic:

sudo ufw allow ssh

sudo ufw status = shows the status of ufw

sudo ufw enable = start the ufw

```
Jan 28 16:19
birat@birat-ubuntu-phd: ~
birat@birat-ubuntu-phd:~$ sudo ufw enable
Firewall is active and enabled on system startup
birat@birat-ubuntu-phd:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)

birat@birat-ubuntu-phd:~$
```

now,

ip a = show the ip address of the computer for ssh connection

```
Jan 28 16:35
birat@birat-ubuntu-phd: ~
birat@birat-ubuntu-phd:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f8:6a:a2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.250.89/24 brd 192.168.250.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86261sec preferred_lft 86261sec
    inet6 fe80::a00:27ff:fe8:6aa2/64 scope link
        valid_lft forever preferred_lft forever
birat@birat-ubuntu-phd:~$
```

now,

go to the cmd of win 11 and type:

ssh birat@192.168.250.89

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\birat>ssh birat@192.168.250.89
The authenticity of host '192.168.250.89 (192.168.250.89)' can't be esta
blished.
ED25519 key fingerprint is SHA256:VmZNeJ0cFf2i+VscAL9hdwGjifugnY4KajrDwF
Ceon0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

in win 11 vm cmd:

ssh-keygen = makes a key pair to connect instead of password

```
C:\WINDOWS\system32\cmd. x + v
C:\Users\birat>ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\birat/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\birat/.ssh/id_ed25519
Your public key has been saved in C:\Users\birat/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:j4Q9N/YqIEFIdvN5XylAJAd1veEfbNHrLn1pPXyvYXA birat@DESKTOP-7C4L1UR
The key's randomart image is:
+--[ED25519 256]--+
| .o.o o=* .. . |
| ....o + o o.. |
| . o . .o+ .. |
| . + . oo +. |
| .. S = .oE. |
| . . * o o.. |
| . . . . * o |
| . . o O+ |
| .. +.* |
+-----[SHA256]-----+

C:\Users\birat>
```

type known_hosts = will show the keys of the known hosts

connect ssh again with the password, then,

vi authorized_keys

then,

```
type id_***.pub
```

then, copy what comes then go back to previous ssh terminal of cmd and then press 'i' to go into insert mode and then press ctrl+shift+v to paste it there.

press esc to exit the insert mode.

then type `:wq` to write it (save) and quit

```
C:\WINDOWS\system32\cmd. x + v
843 known_hosts
01/28/2026 01:36 PM
01/28/2026 01:36 PM
C:\Users\birat\ ssh-ed25519 AAAA...
The system cannot find the file specified.
C:\Users\birat\ C:\Users\birat\.ssh>ssh birat@192.168.250.89
birat@192.168.250.89's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

11 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Wed Jan 28 16:37:27 2026 from 192.168.250.163
birat@birat-ubuntu-phd:~$ cd .ssh
birat@birat-ubuntu-phd:~/..ssh$ vi authorized_keys
birat@birat-ubuntu-phd:~/..ssh$ cat authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAICwMUIId9WMZ6XunLJ2kXp2QycCCofSRG7YMPbQQ1PFRr birat@DESKTOP-7C4L1UR
birat@birat-ubuntu-phd:~/..ssh$
```

now it just connects, does not ask for the password as we already have the key.

```
0026 01:36 PM 843 known_hosts
0026 01:
rs\birat\
05519 AAA
rs\birat\
stem cann
rs\birat\
e in driv
e Serial
ctory of C
0026 01:
0026 01:
0026 01:
0026 01:
0026 01:
0026 01:
rs\birat\
05519 AAA
rs\birat\
Last login: Wed Jan 28 16:44:36 2026 from 192.168.250.163
birat@birat-ubuntu-phd:~$ |
```

RDP = Remote Desktop Protocol

now, making the win 11 vm as remote desktop server, and install remmina in ubuntu and connect to the server.

allow remote desktop setting in win 11

get the ip address from the cmd of win 11 with ipconfig.

```
C:\WINDOWS\system32\cmd. X + -
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\birat>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : home.local
    Link-local IPv6 Address . . . . . : fe80::ba7d:c3aa:f1ae:a2a1%10
    IPv4 Address. . . . . : 192.168.250.163
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.250.1

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Unknown adapter OpenVPN Connect DCO Adapter:

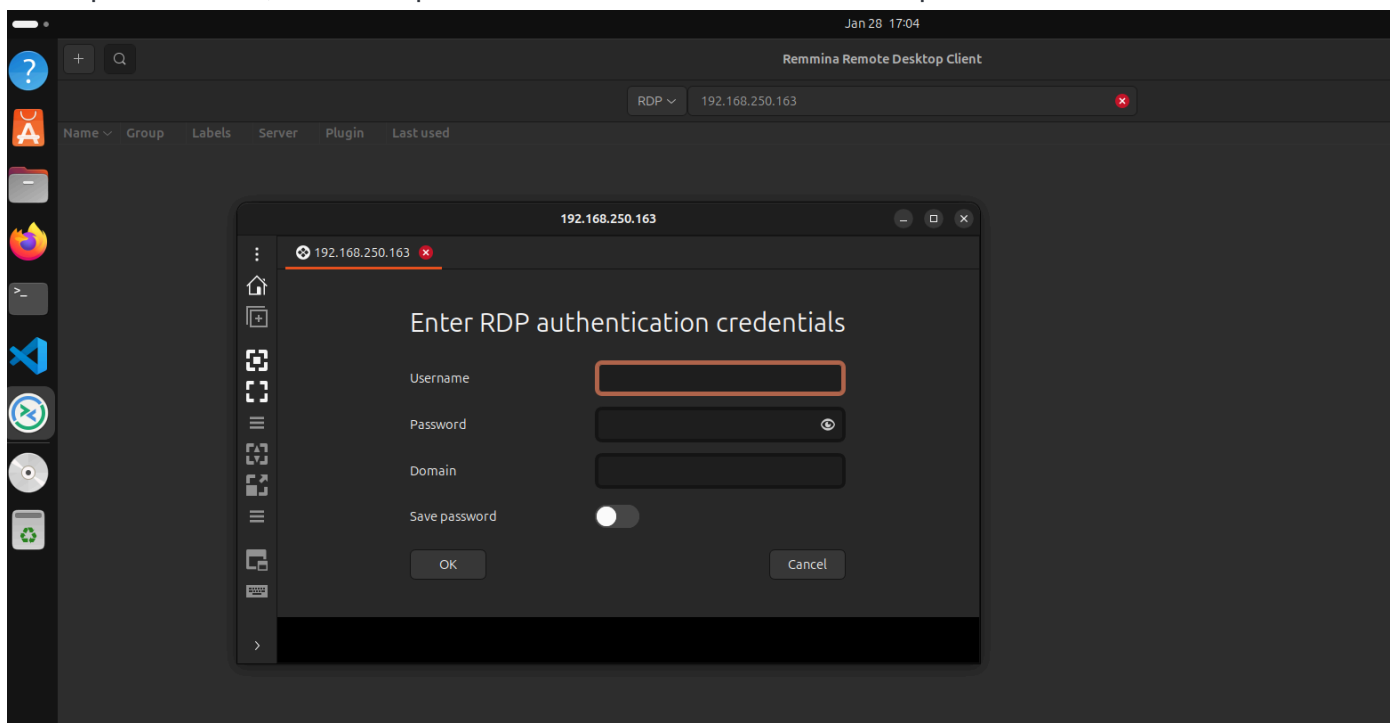
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\Users\birat>
```

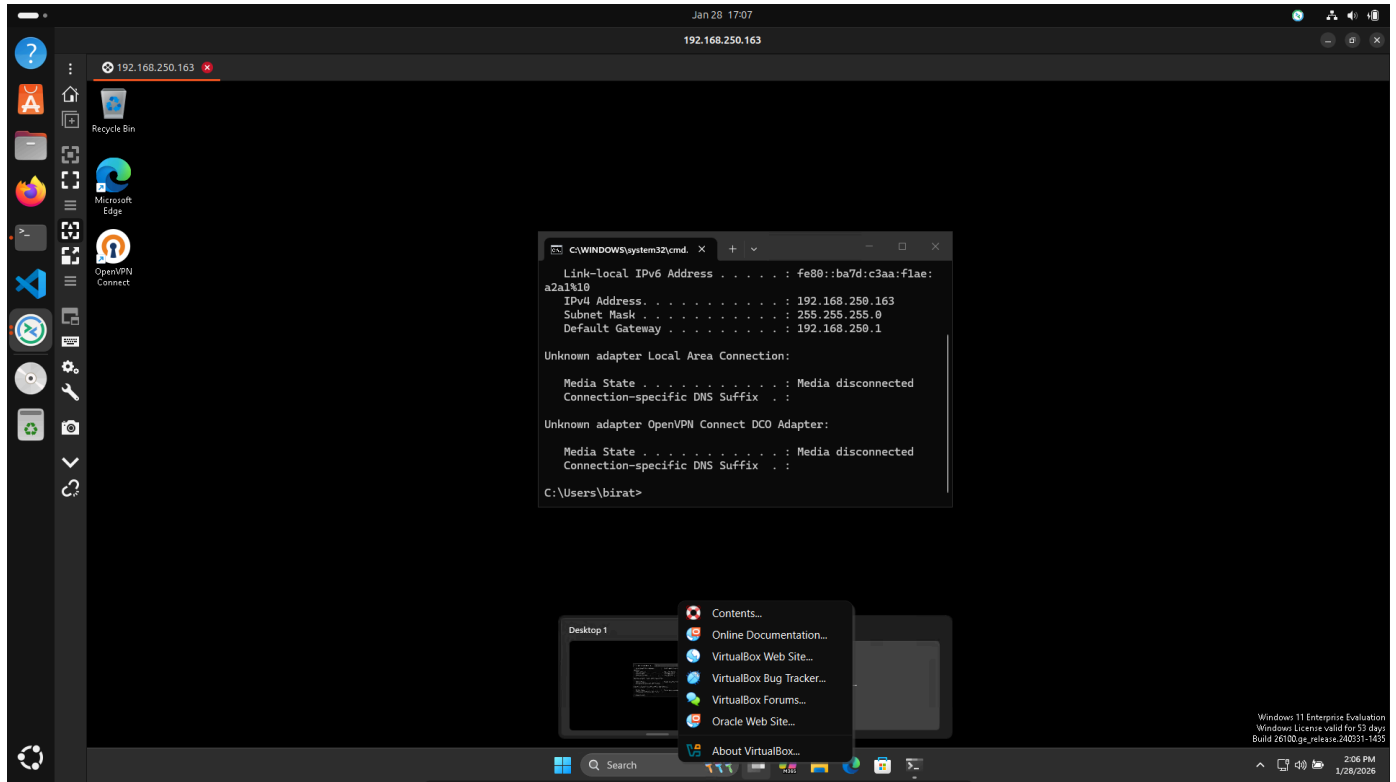
now, in ubuntu:

sudo apt install remmina

then open remmina, enter the ip address of win 11 vm and then accept the certificate.



toggle it to full screen after maximizing the window.

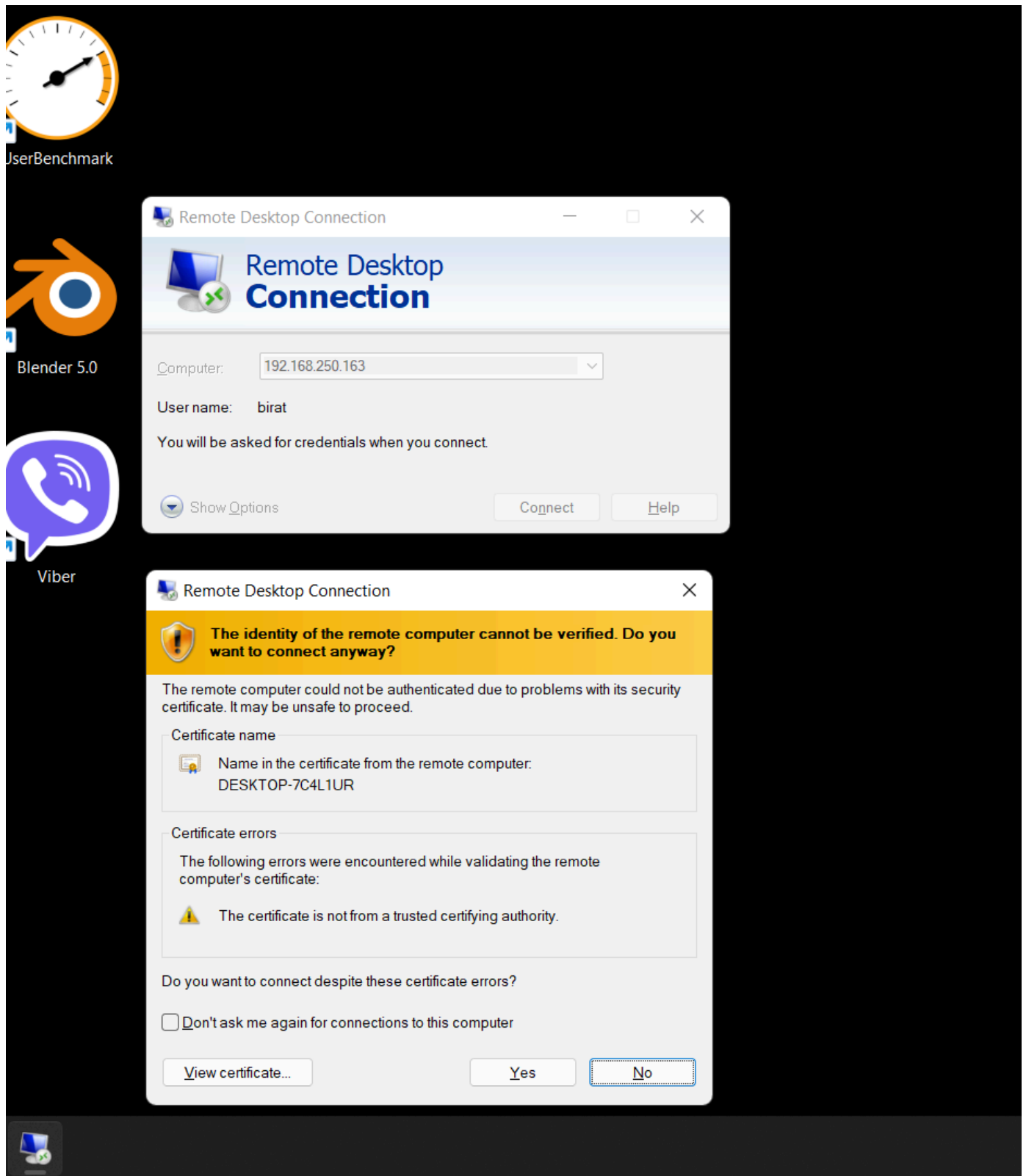


now in windows to run remote desktop client:

win + r = open run command

then type:

mstsc

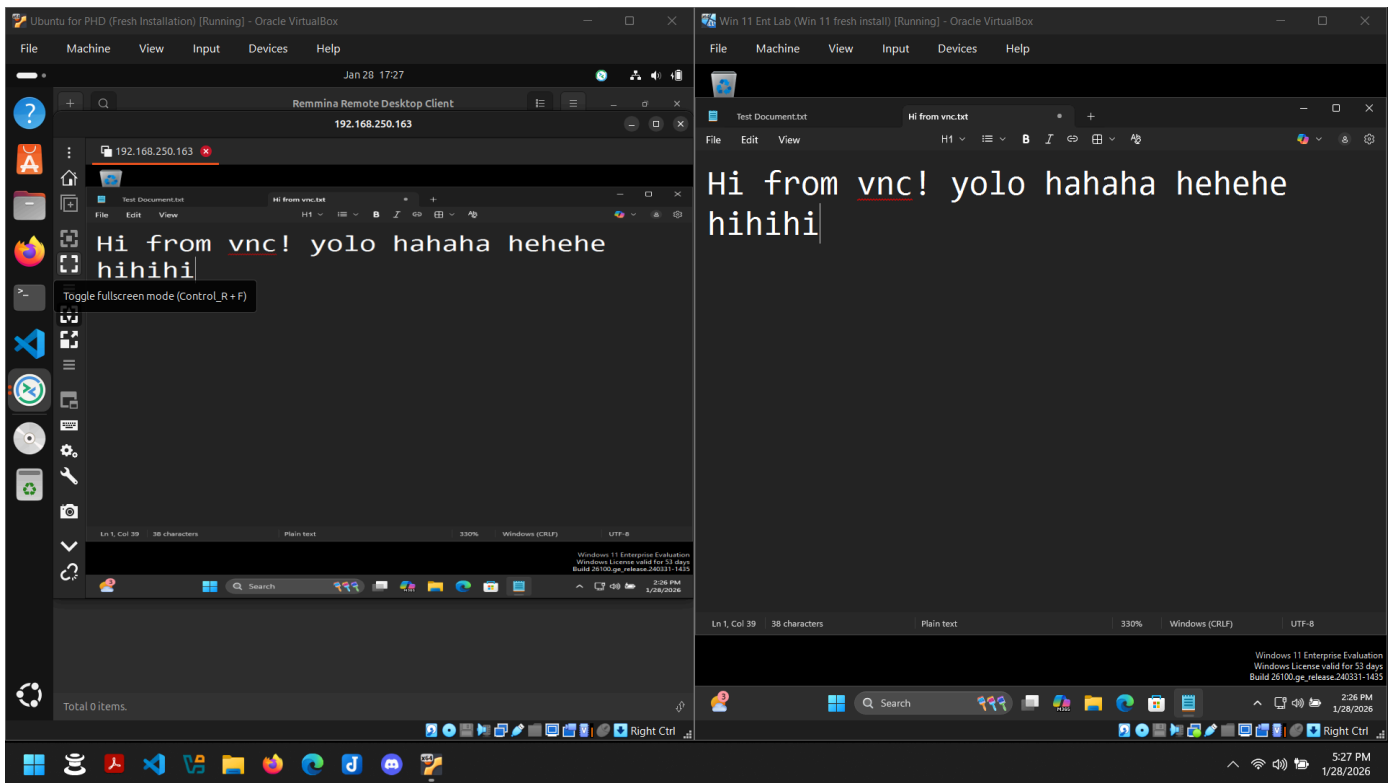


accept the certificate.

VNC: Virtual Network Computing

download and install tightvnc on win 11 vm and then set password for it. it will be on system tray after you press ok.

then, open remmina on ubuntu vm and then select vnc from dropdown and then put in the ip address of the win 11 vm and then put the password.

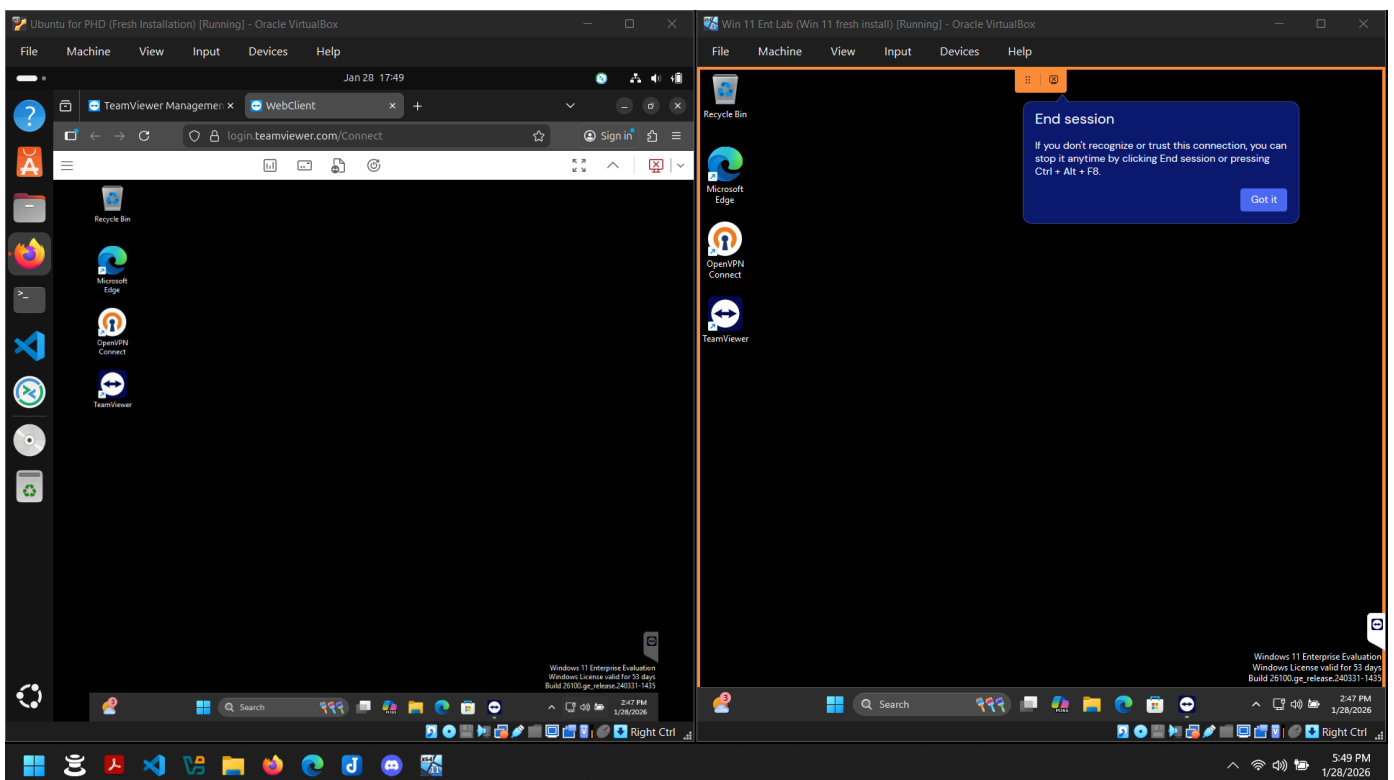


vnc should not be exposed to the internet, but can be done using vpn. it updates in real time for the client. while ssh and rdp would log you out from all the users in the client while working from the server.

third party softwares for remote support:

anydesk, teamviewer, logmeandrescue, gotomypc, etc...

teamviewer also is like vnc, it lets the client use the computer while remote support is going on.



Tips and tricks to provide phone support:

1. Provide Clear and Concise Instructions
2. Avoid Jargon or Acronyms
3. Follow along on your Computer
4. Avoid Skipping Steps